ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Area & Perimeter

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

AREA & PERIMETER · CH4 · L5

ENG. ESLAM AHMED

The Map

This lesson collapses Area & Perimeter into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Split the compound shape
TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Build an equation from area
TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use perimeter after missing sides
TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check units squared or linear
TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient
TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 32 website questions, 32 checked solutions, 4 local PDFs read.

01 Split the compound shape

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$12 \times 9 = 108 \text{ cm}^2$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q3

Split the shape into two
This is a reasoning step: write one clear sentence, then continue the calculation.

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

Finish: 7 cm.

FORMULA BANK

Break the shape. Rectangle lw, triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 3 The diagram shows a shape. 9 cm x cm 6 cm 12 cm Diagram NOT accurately drawn The shape has area 129 cm2 Work out the value of x. x =

YOUR SOLUTION

02 Build an equation from area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = \frac{1}{2}\pi r^2$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q5
The garden is a semicircle

$$\text{area} = \frac{1}{2}\pi r^2$$

Calculate area

$$12 \times 6 = 72 \text{ m}^2$$

Finish: No, he does not have enough grass seed.

FORMULA BANK

Break the shape. Rectangle lw, triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 5 The diagram shows Yuen's garden. 7.2 m Diagram NOT accurately drawn The garden is in the shape of a semicircle of radius 7.2 m. Yuen is going to cover his garden with grass seed. Yuen has 12 boxes of grass seed. Each box of grass seed contains enough seed to cover 6 m2 of the garden. Has Yuen enough grass seed for his garden? Show your working clearly.

YOUR SOLUTION

03 Use perimeter after missing sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$12 \times 9 = 108 \text{ cm}^2$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q3

Split the shape into two
This is a reasoning step: write one clear sentence, then continue the calculation.

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

Finish: 7 cm.

FORMULA BANK

Break the shape. Rectangle lw, triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 3 The diagram shows a shape. 9 cm x cm 6 cm 12 cm Diagram NOT accurately drawn The shape has area 129 cm2 Work out the value of x. x =

YOUR SOLUTION

04 Check units squared or linear

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AD = 17\text{ cm}, \qquad CD = 15\text{ cm}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q11

Use trigonometry

$$AD = 17\text{ cm}, \qquad CD = 15\text{ cm}$$

Use Pythagoras to find

$$AC^2 = 17^2 - 15^2 = 289 - 225 = 64$$

Finish: 44.6 cm.

FORMULA BANK

Break the shape. Rectangle lw, triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 11 The diagram shows a shaded shape ABCD made from a semicircle ABC and a right-angled triangle ACD. Diagram NOT accurately drawn 17 cm 15 cm A B C D AC is the diameter of the semicircle ABC. Work out the perimeter of the shaded shape. Give your answer correct to 3 significant figures. cm

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$PQ = p$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q20

Let

$$PQ = p$$

Given

$$AB = 1.5PQ$$

Finish: $PQ = 4.8$ cm

FORMULA BANK

Break the shape. Rectangle lw, triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 20 A B F P Q C D R E Diagram NOT accurately drawn T The diagram shows a shaded region T formed by removing an equilateral triangle PQR from a regular hexagon ABCDEF. The points P and Q lie on AB such that AB = 1.5 times PQ Given that the area of region T is 72 3 cm2 work out the length of PQ. cm

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Split the compound shape
"work out", "show", "hence", a familiar exam phrase	Build an equation from area
"work out", "show", "hence", a familiar exam phrase	Use perimeter after missing sides
"work out", "show", "hence", a familiar exam phrase	Check units squared or linear
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.